

Unidad 2

Redes y Transmisión de datos II ALTD

- Función de dispositivos de capa 2,
- BRIDGE SWITCH
- PLANTEAR TOPOLOGIAS Y CASCADAS EN PACKET TRACER
- CONFIGURAR COMPONENTES-

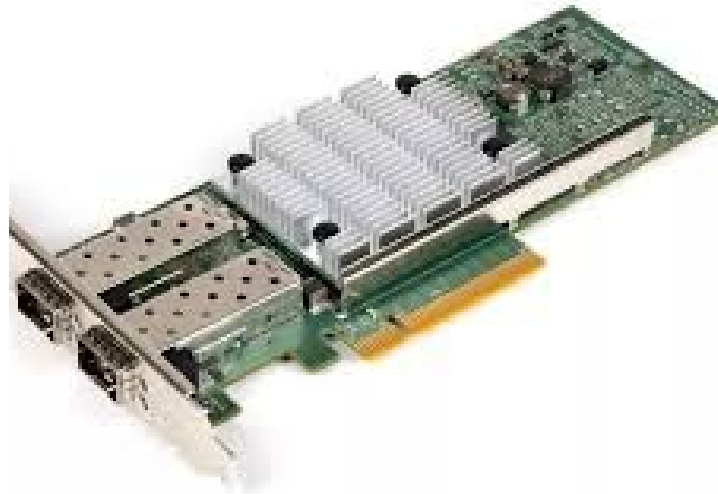
switch

Enlazar datos de capas superiores a la red

- NIC: Conexión de dispositivo de red



802.3 Ethernet

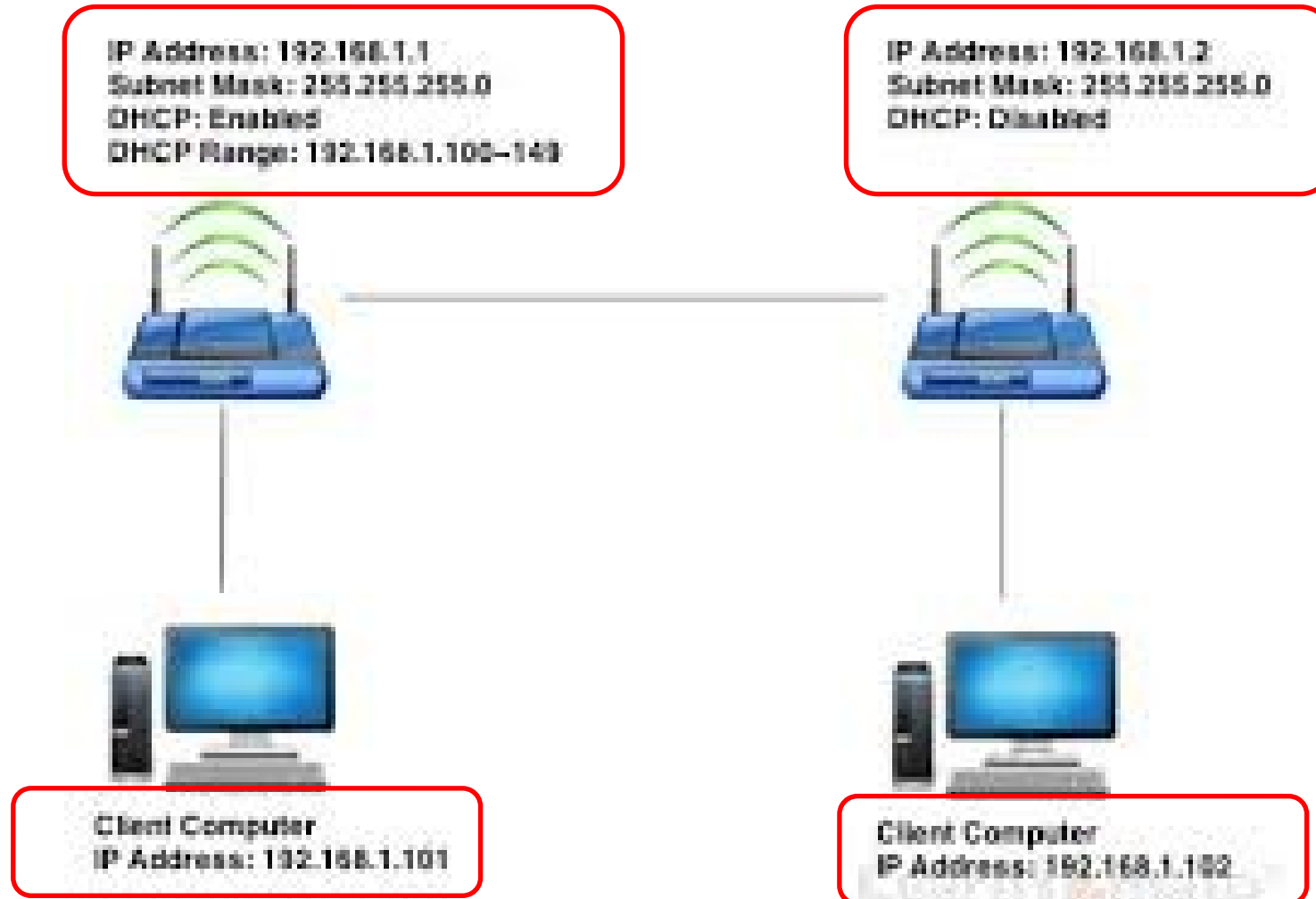


802.8 F.O.

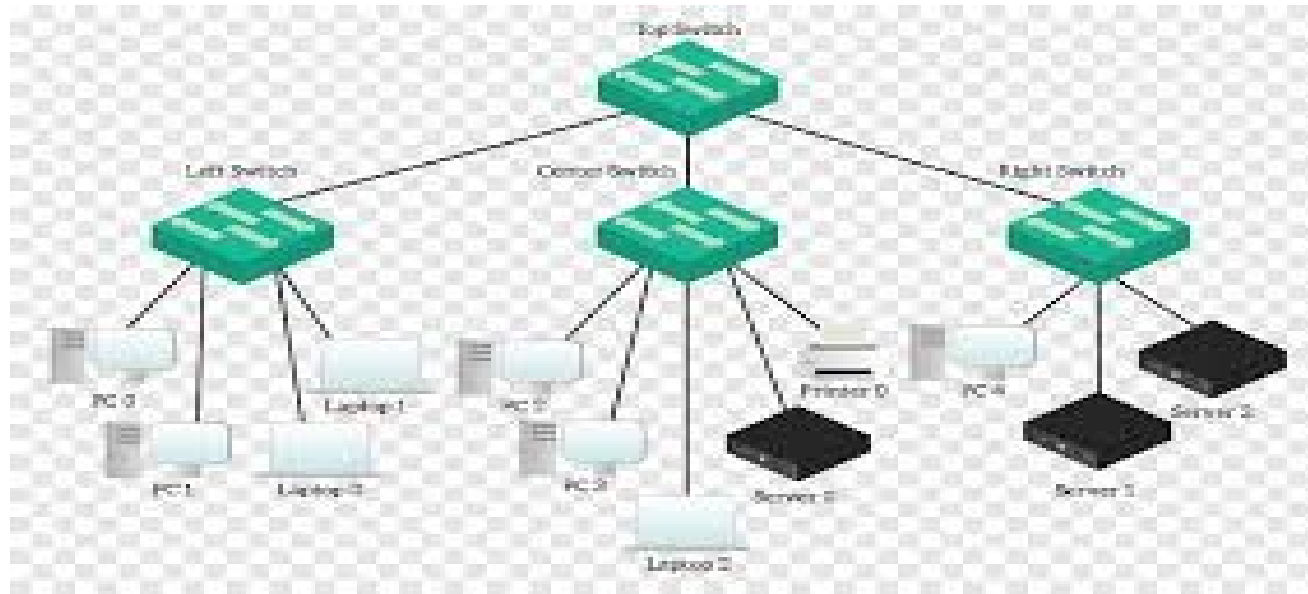
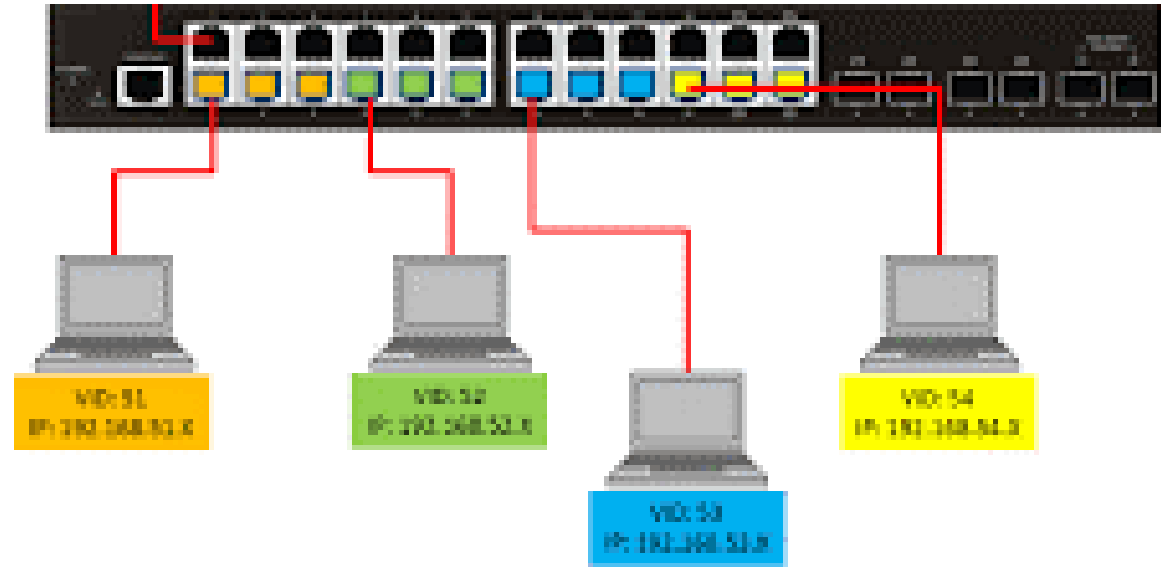


802.11 WiFi

BRIDGE: extiende el alcance LAN

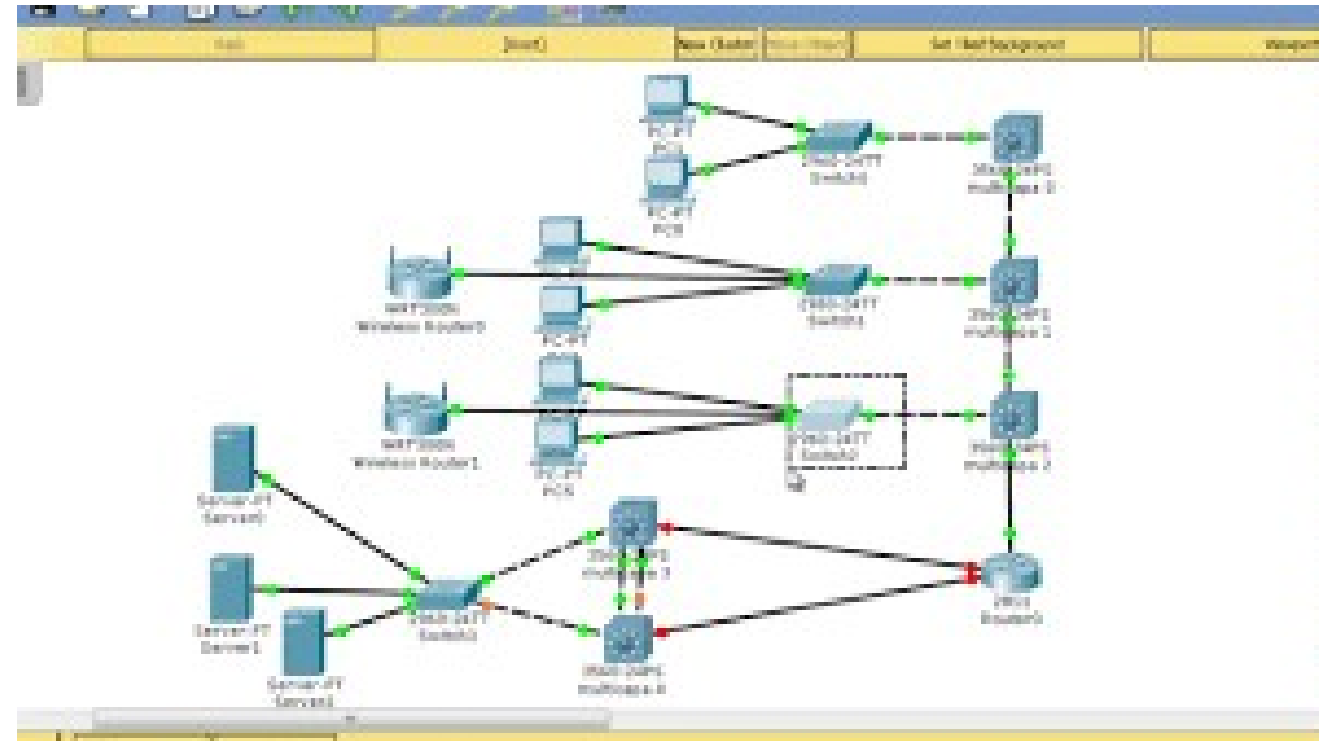
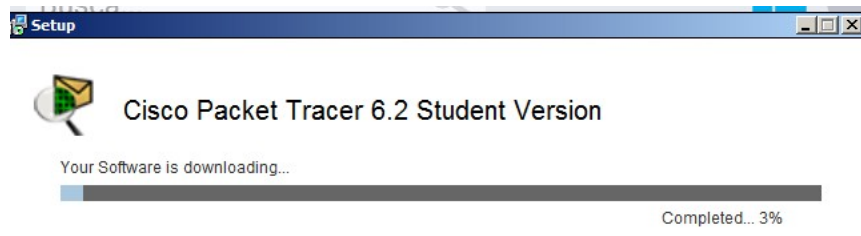


SWITCH: Distribuir trafico por dirección MAC



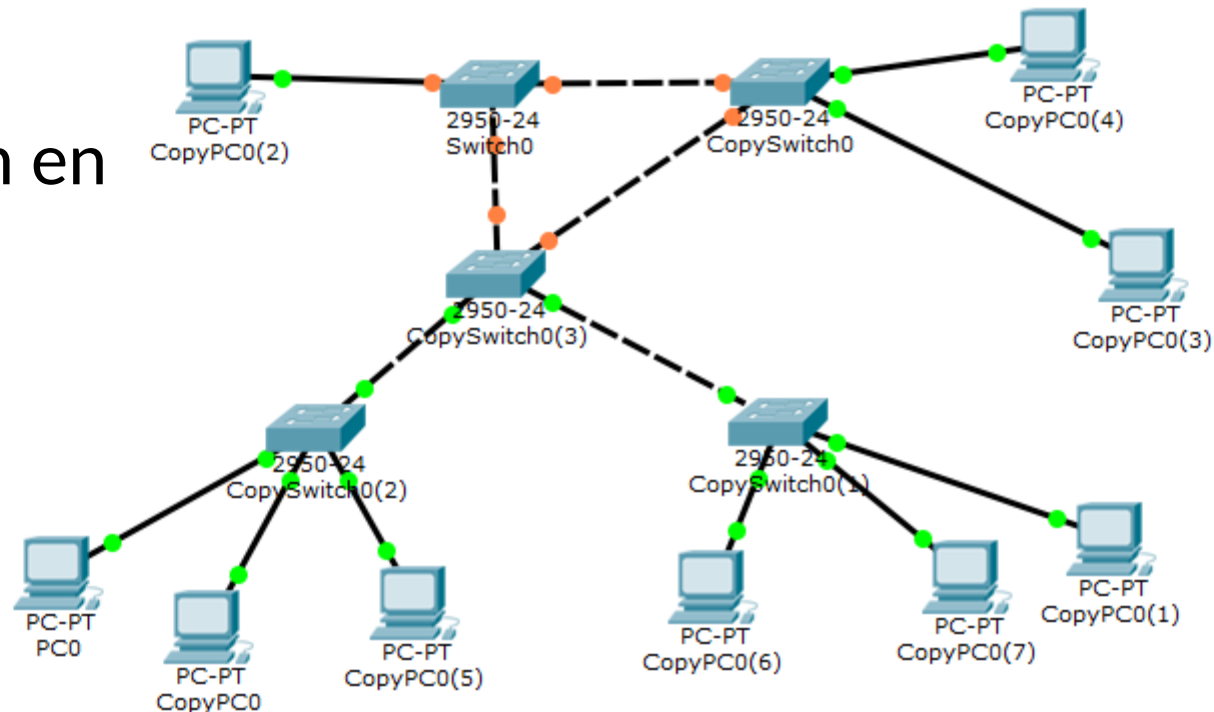
PLANTEAR TOPOLOGIAS Y CASCADAS EN PACKET TRACER (Por norma no mas de 4 cascadas)

- Deberemos descargar e instalar [packet tracer 6.22](#) y configurar la mayor cantidad de topologías configurando las terminales



CONFIGURAR COMPONENTES

- [Manual básico packet tracer](#)
- [Configuración básica](#), para cada switch
- Tarea 1, identificar utilidad de comandos en configuración básica
- Tarea 2, implementar al menos 3 topológicas distintas con switch en cascada como la figura
- Considerar 1 LAN
- Salvar cada configuración en block de notas

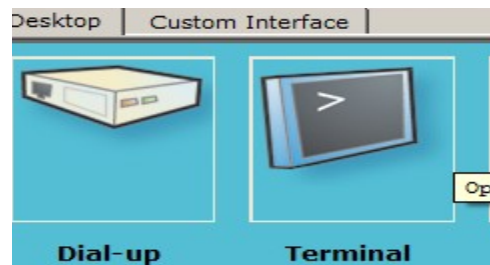


conectand



1 conectar cable terminal rs232

2 Abrir terminal pc



Terminal Configuration

Port Configuration

Bits Per Second: 9600
Data Bits: 8
Parity: None
Stop Bits: 1
Flow Control: None

OK

Terminal

Press RETURN to get started!

Prohibido el acceso a este equipo

User Access Verification

Password:

SW_3>ena

Password:

SW_3#sh

SW_3#show mac

SW_3#show mac ad

SW_3#show mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	000c.cf48.bb02	DYNAMIC	Fa0/3
1	00e0.f711.e902	DYNAMIC	Fa0/2

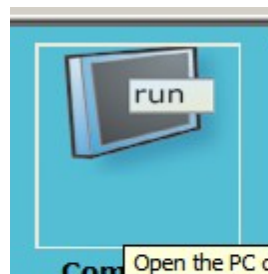
SW_3#

3 cargar la configuración básica editando pass
hostname e ip de cada equipo

La tecla tab completa comandos

La tecla ? despliega ayuda , verificar con comandos
show etc,

4 Verificar con comandos de red: ping, tracer
ipconfig arp -a en pantalla de comandos del PC



```
enable
```

```
configure terminal
```

```
hostname SW_3
```

```
enable password cisco123
```

```
enable secret cisco123
```

```
line console 0
```

```
password cisco123
```

```
login
```

```
exit
```

```
line VTY 0 4
```

```
password cisco123
```

```
login
```

```
exit
```

```
service password-encrypt
```

```
no ip domain-lookup
```

```
banner motd # Prohibited
```

```
interface vlan 1
```

```
ip address 192.168.1.1
```

```
no shutdown
```

```
ip default-gateway 192.168.1.1
```

```
exit
```

```
copy running-config startup
```

The screenshot displays the Cisco Packet Tracer Student application. On the left, a 'Logical' network diagram shows a switch and a PC connected. The PC is selected, and its configuration window is open. The 'Config' tab is active, showing the PC's settings. A 'Command Prompt' window is overlaid on the PC configuration, showing the output of several commands. The commands and their outputs are as follows:

```
PC>ipconfig

FastEthernet0 Connection:(default port)

    Link-local IPv6 Address.....: FE80::2D0:97FF:FEED:5157
    IP Address.....: 192.168.1.2
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: 0.0.0.0

PC>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=186ms TTL=128
Reply from 192.168.1.1: bytes=32 time=0ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 186ms, Average = 46ms

PC>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:
```

The bottom status bar shows 'Time: 00:22' and 'Conne'.

Se debe
practicar
mucho